

Activity 1 – Loading Dataset and Visualization

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Using KNIME Analytics Platform

Aim

The aim of this activity is to load a dataset into KNIME Analytics Platform and to explore it using various visualization techniques such as table views, histograms, scatter plots, box plots, bar charts, and pie charts, in order to understand the structure, distribution, and relationships within the data.

Introduction

KNIME (Konstanz Information Miner) is an open-source data analytics platform that allows users to build visual workflows using drag-and-drop nodes, without requiring extensive coding. Loading a dataset is typically the first step in any data analysis workflow, achieved through nodes such as the File Reader or CSV Reader, which import structured data from external sources into KNIME. Once the data is loaded, visualization plays a critical role in understanding its underlying patterns, distributions, and relationships. KNIME provides a wide range of visualization nodes, including Table View, Histogram, Scatter Plot, Box Plot, Bar Chart, and Pie Chart, each suited to different types of analysis. Together, dataset loading and visualization form the foundation for exploratory data analysis (EDA), enabling analysts to identify trends, detect outliers, and make informed decisions before proceeding to advanced modeling tasks.

KNIME Workflow

The overall workflow created for this activity consists of a data-loading node connected to multiple visualization nodes.

Visualization Results

This section presents the visual outputs generated at each stage of the workflow, from the raw dataset to the various visualization charts.

1. Dataset Loaded

The screenshot displays the KNIME Analytics Platform interface. A pop-up window in the top-left corner provides instructions: "Add and connect nodes to perform actions on data". The main workspace shows an "Excel Reader" node configured with the following settings:

- File and Sheet:**
 - Mode: File
 - Source: D:\AI Course\Karnataka_APMC_Mandi_Prices.xlsx
 - Select sheet: First sheet with data
 - Sheet name: APMC Mandi Prices
- Data Area:** Discard, Apply, Apply and Execute

The bottom of the interface shows the "Table" view of the data. The table has 2176 rows and 10 columns. The first four rows are visible:

#	RowID	Karnataka ...	empty_B	empty_C	empty_D	empty_E	empty_F	empty_G
1	Row0	Date	District	Market HQ	APMC / Mandi	Commodity	Min Price (₹/...	Max Price (₹/...
2	Row1	29-03-2025	Bengaluru Urb...	Bengaluru	Yeshwanthpu...	Maize	1620.94	1991.62
3	Row2	29-03-2025	Bengaluru Urb...	Bengaluru	Yeshwanthpu...	Paddy (Com...	1452.47	1694.19
4	Row3	29-03-2025	Bengaluru Urb...	Bengaluru	Yeshwanthpu...	Sunflower	4632.26	5303.68

Figure 2: Dataset successfully loaded into KNIME

2. Table View

► 1: File Table Flow Variables

Rows: 150 | Columns: 5 Sorting Filter Table Statistics Search

#	RowID	sepal length (cm) Number (Float)	sepal width (cm) Number (Float)	petal length (cm) Number (Float)	petal width (cm) Number (Float)	target String
1	Row0	5.1	3.5	1.4	0.2	Iris-setosa
2	Row1	4.9	3	1.4	0.2	Iris-setosa
3	Row2	4.7	3.2	1.3	0.2	Iris-setosa
4	Row3	4.6	3.1	1.5	0.2	Iris-setosa
5	Row4	5	3.6	1.4	0.2	Iris-setosa
6	Row5	5.4	3.9	1.7	0.4	Iris-setosa
7	Row6	4.6	3.4	1.4	0.3	Iris-setosa
8	Row7	5	3.4	1.5	0.2	Iris-setosa
9	Row8	4.4	2.9	1.4	0.2	Iris-setosa
10	Row9	4.9	3.1	1.5	0.1	Iris-setosa
11	Row10	5.4	3.7	1.5	0.2	Iris-setosa
12	Row11	4.8	3.4	1.6	0.2	Iris-setosa
13	Row12	4.8	3	1.4	0.1	Iris-setosa
14	Row13	4.3	3	1.1	0.1	Iris-setosa
15	Row14	5.8	4	1.2	0.2	Iris-setosa

Figure 3: Tabular view of the dataset

3. Histogram

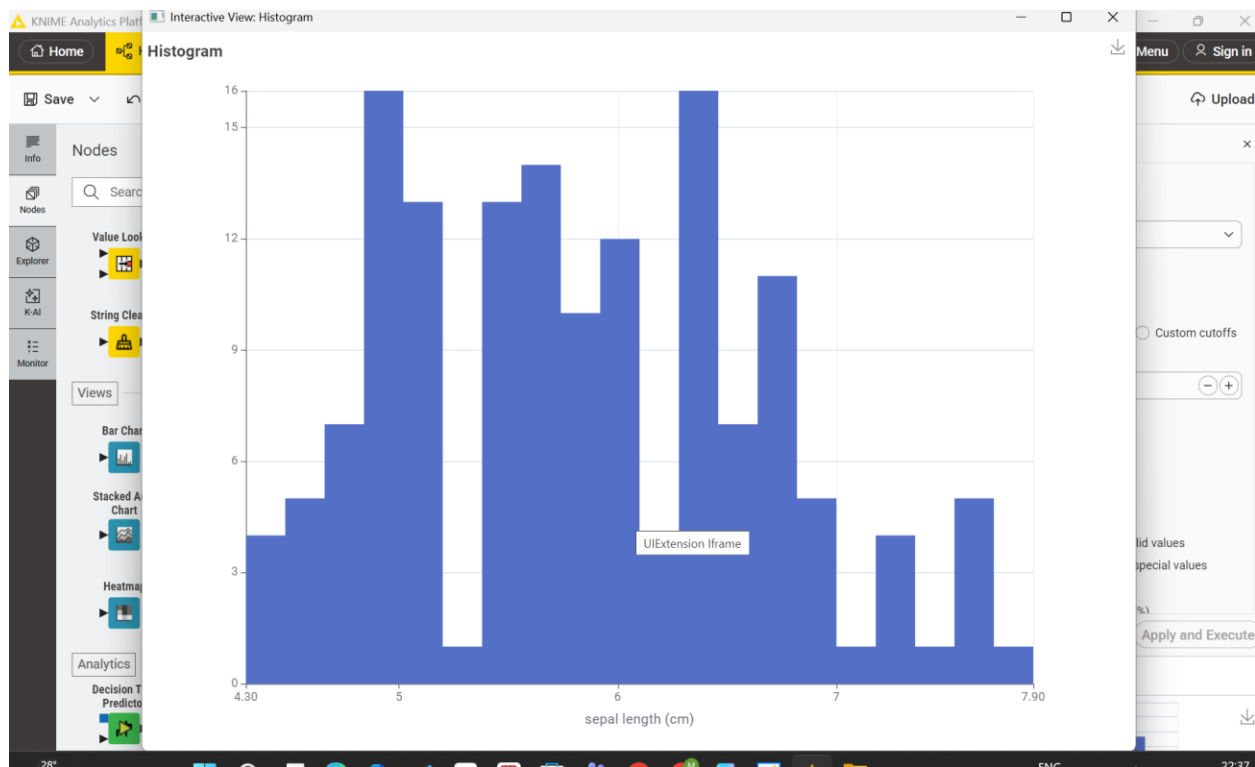


Figure 4: Histogram showing the distribution of a selected attribute

4. Scatter Plot

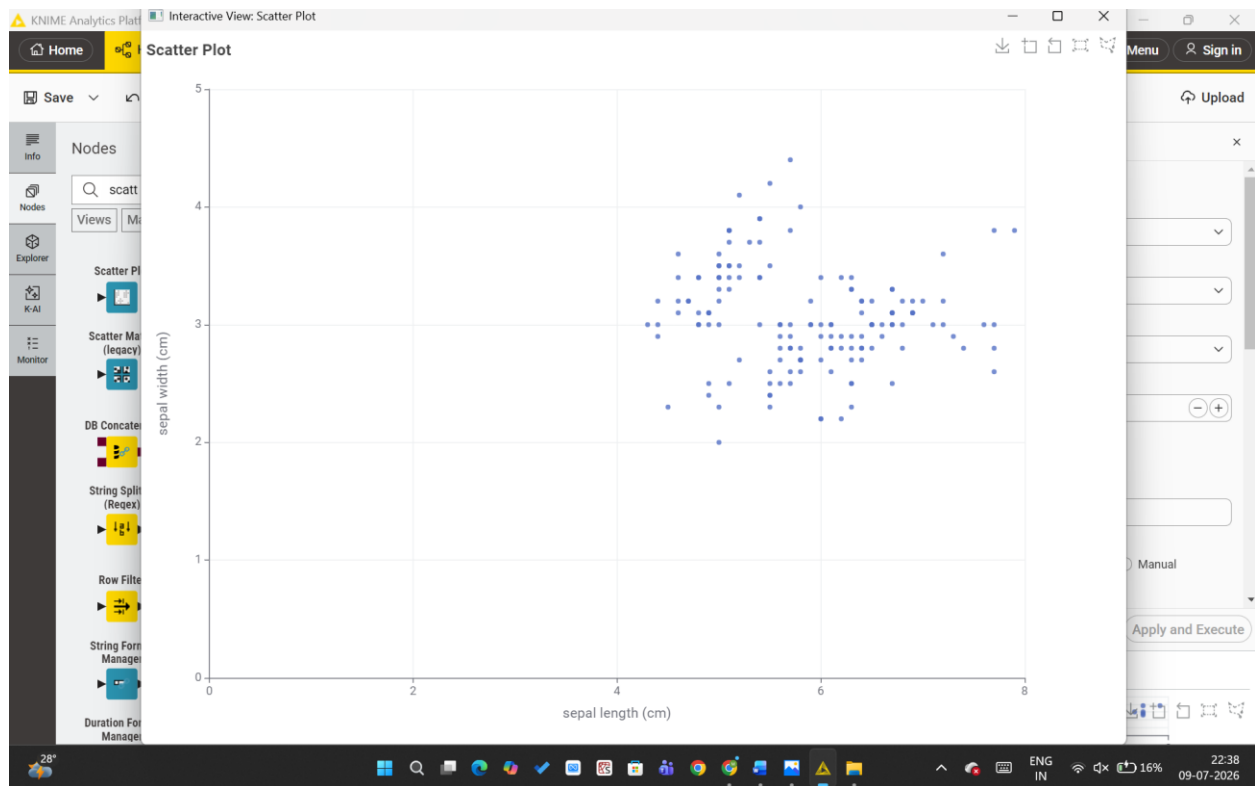


Figure 5: Scatter plot showing relationship between two attributes

5. Box Plot

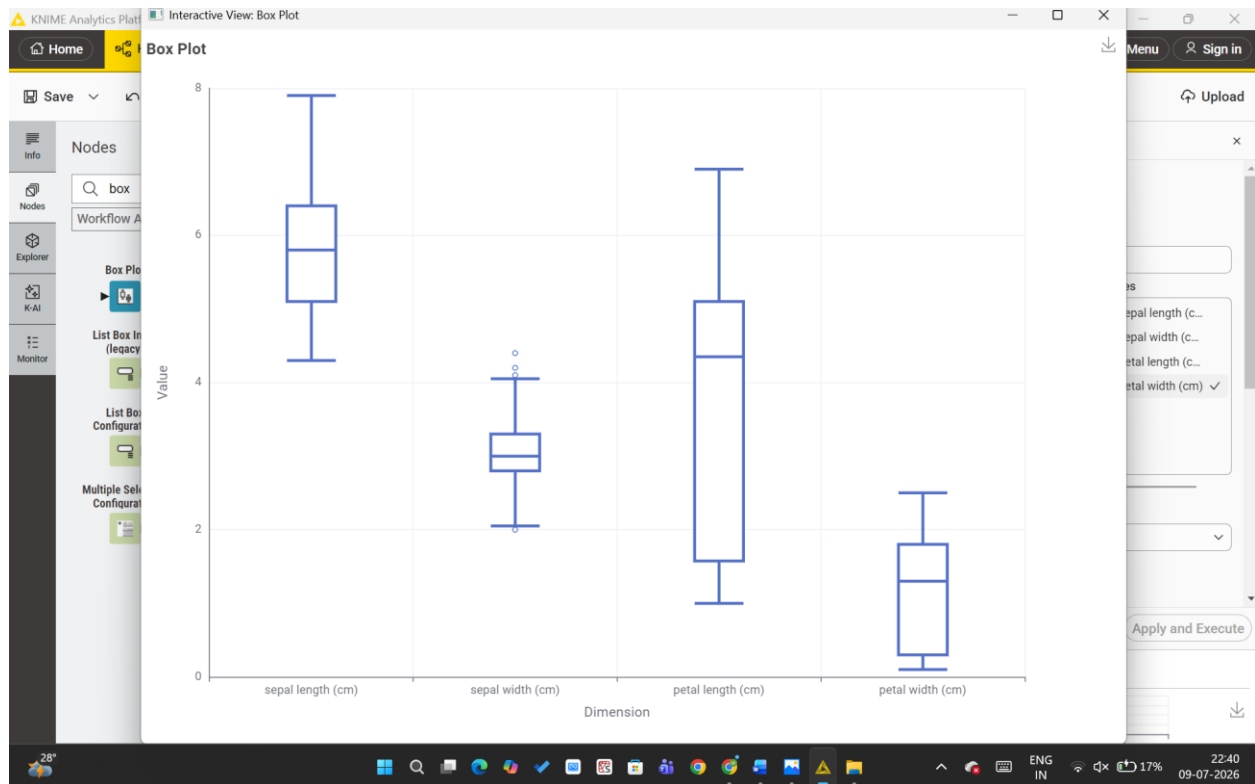


Figure 6: Box plot showing spread and outliers in the data

6. Bar Chart (if used)

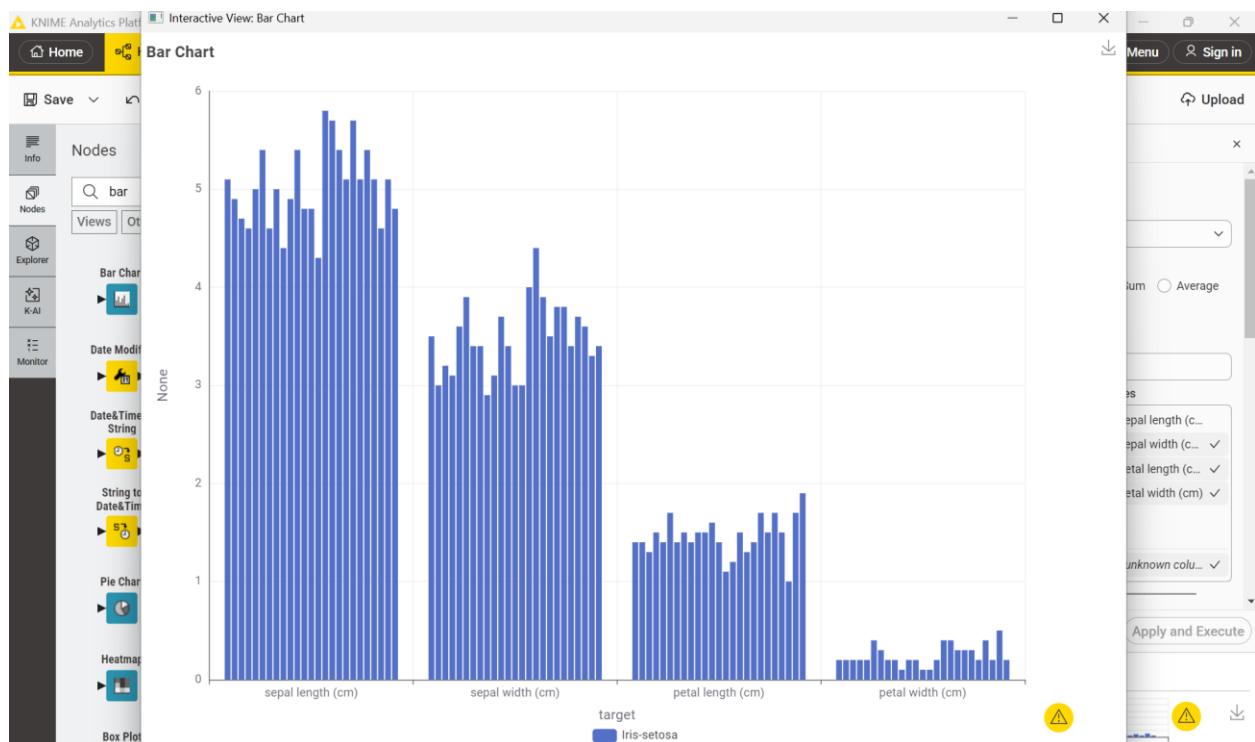


Figure 7: Bar chart showing category-wise comparison

7. Pie Chart (if used)

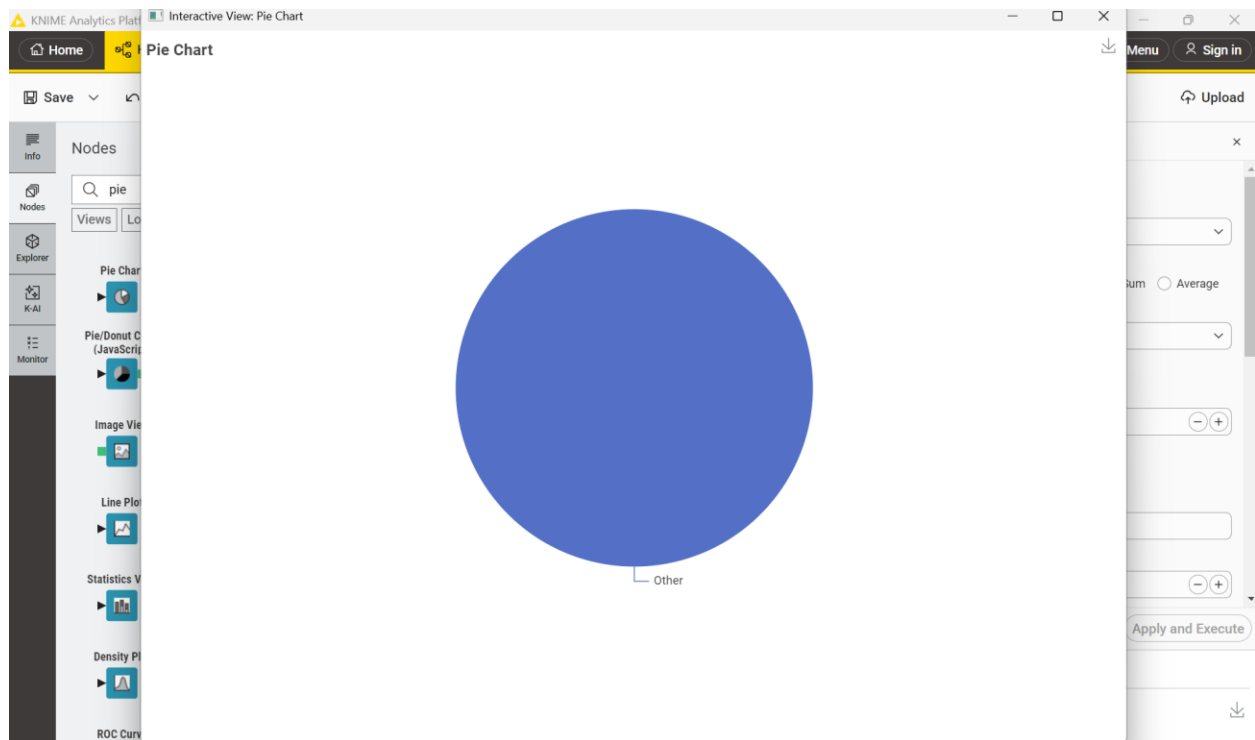


Figure 8: Pie chart showing proportional distribution of categories

KNIME Nodes Used

The following table briefly describes each node used in the workflow:

Node	Description
File Reader / CSV Reader	Imports a dataset from a local file (such as .csv) into the KNIME workflow, converting it into a KNIME data table for further processing.
Table View	Displays the loaded dataset in a simple row-and-column tabular format, allowing users to inspect raw data values directly.
Histogram	Plots the frequency distribution of a selected numeric attribute, helping to visualize how values are spread across different ranges.
Scatter Plot	Plots data points on a two-dimensional graph using two selected attributes, useful for identifying correlations or clusters.
Color Manager	Assigns distinct colors to categorical values in the dataset, making it easier to differentiate groups across visualizations.

Node	Description
Box Plot	Displays the median, quartiles, and outliers of numeric attributes, summarizing the statistical spread of the data.
Pie Chart	Represents categorical data as proportional slices of a circle, useful for showing relative percentages of categories.
Bar Chart	Compares numeric values across different categories using rectangular bars, useful for category-wise analysis.

Workflow Explanation

Step 1: Load the Dataset — The File Reader or CSV Reader node is added to the workflow canvas and configured with the path of the dataset file. Executing this node imports the data into KNIME.

Step 2: Inspect the Data — A Table View node is connected to the reader node to display the dataset in tabular form, allowing verification that the data has loaded correctly.

Step 3: Generate Visualizations — Visualization nodes such as Histogram, Scatter Plot, Box Plot, Bar Chart, and Pie Chart are connected to the data table to explore different aspects of the dataset.

Step 4: Apply Color Coding — The Color Manager node is used, where required, to assign colors to categorical attributes so that groups are visually distinguishable across charts.

Step 5: Execute and View Results — Each node is executed in sequence, and the output views are opened to observe the generated charts and interpret the results.

Result

The dataset was successfully loaded into KNIME and visualized using multiple chart types, including table view, histogram, scatter plot, box plot, bar chart, and pie chart. These visualizations provided clear insights into the distribution, relationships, and categorical proportions present in the dataset, confirming that the workflow was correctly configured and executed.

Conclusion

Loading and visualizing datasets are essential first steps in any data analysis process, as they help establish a clear understanding of the data before deeper analysis or modeling begins. KNIME simplifies this process through its intuitive, node-based interface, eliminating the need for manual coding. Visualization techniques such as histograms, scatter plots, and box plots reveal patterns, trends, and anomalies that might otherwise remain hidden in raw data. This activity demonstrated

how effectively KNIME can be used to import, explore, and visually represent data, laying a strong foundation for subsequent data preprocessing and machine learning tasks.